

PAPER • OPEN ACCESS

Mission Design for Multi-Target Jupiter System Exploration and Ring Sample Return Using Gravity Assists

To cite this article: Yuxia Wang *et al* 2025 *J. Phys.: Conf. Ser.* **3109** 012085

View the [article online](#) for updates and enhancements.

You may also like

- [Earths Are Not Super-Earths, Saturns Are Not Jupiters: Imprints of Pressure-bump Planet Formation on Planetary Architectures](#)
Wenrui Xu, and Songhu Wang
- [The fundamental concepts of the gravity-assist manoeuvre](#)
P Berg
- [Exact maps in density functional theory for lattice models](#)
Tanja Dimitrov, Heiko Appel, Johanna I Fuks et al.

Mission Design for Multi-Target Jupiter System Exploration and Ring Sample Return Using Gravity Assists

Yuxia Wang¹, Shaoyang Lyu^{1, 2} and Xiangyu Li^{1, 3*}

¹School of Aerospace Engineering, Beijing Institute of Technology, 100081 Beijing, People's Republic of China

²Key Laboratory of Dynamics and Control of Flight Vehicle of Ministry of Education, Beijing, 100081, China

³Department of Strategic and Advanced Interdisciplinary Research, Peng Cheng Laboratory, Shenzhen, Guangdong Province, 518055, China

*E-mail: lixiangy@bit.edu.cn

Abstract. This paper presents a trajectory design framework for a multi-target exploration mission within the Jovian system, incorporating the Gossamer ring sample return and Jovian moon exploration. First, the interplanetary transfer from Earth to Jupiter is analyzed using Tisserand diagrams to screen gravity assist sequences, with further optimization conducted via the MGA method. The EVEEJ (Earth–Venus–Earth–Earth–Jupiter) sequence is identified as the optimal transfer path. Next, the spacecraft crosses Jupiter's Gossamer ring twice during the close Jupiter flyby, enabling dust sample collection using aerogel-based capture technology. The main probe carrying the return capsule transfers into the Earth-bound trajectory using the JEEME (Jupiter–Earth–Earth–Mars–Earth) return sequence. Furthermore, a sub-probe is deployed in the Jovian system to conduct extended scientific investigations. An intra-Jupiter system exploration orbit is designed, including Jupiter capture, intra-system gravity assist transfers, and Callisto orbit insertion. The detailed mission profile is designed and analyzed. The sample return mission requires a total velocity increment of 0.3329 km/s with a duration of 14.8 years, and the total ΔV for the intra-system mission is 1.8212 km/s over 818 days, which is available based on the current spacecraft capability. The proposed scheme demonstrates a balanced approach between engineering feasibility and scientific value, offering a viable solution for complex outer planetary missions.

1. Introduction

Jupiter, the largest gas giant in the solar system, exhibits the fastest rotational velocity and the most powerful magnetic field among all planets. It hosts a complex planetary system composed of 79 known natural satellites. Notably, the Galilean moons (Io, Europa, Ganymede, and Callisto) have drawn significant scientific interest due to their planet-like scales and the potential existence of subsurface oceans. The geological activity and possible habitability of icy moons such as Europa and Ganymede offer valuable insights into planetary evolution and the prospects of extraterrestrial life.

Since the 1970s, some international missions have targeted the Jovian system through flybys, orbital insertions, and atmospheric entry maneuvers. Pioneer 10 achieved the first Jupiter flyby and revealed fundamental characteristics of its magnetic field and radiation environment^[1].

Galileo entered Jupiter's atmosphere and provided strong evidence for the existence of subsurface oceans on its moons^[2]. Juno adopted a polar orbit to extend mission duration and mitigate radiation exposure^[3].

Despite multiple exploration efforts, Jupiter remains a challenging target due to long transit times and its intense radiation environment. To maximize scientific return, this study proposes an innovative dual-probe mission program for comprehensively exploring the Jupiter system, balancing scientific objectives with engineering feasibility.

To date, deep space sample return missions have been confined to the Moon, asteroids, and comets, with no prior missions successfully retrieving material from the outer planetary systems, such as Jupiter. Establishing sample return as the primary mission objective greatly enhances both the scientific depth and overall impact of the mission. Unlike remote sensing or in situ analysis, sample return enables high-precision laboratory investigations of well-preserved samples, providing physical and chemical information of a quality that is otherwise difficult to obtain in situ. Such data are critical for advancing our understanding of the origin and evolution of the solar system.

This paper presents a complete trajectory design process from interplanetary transfer to Jovian system operations. The mission is divided into two components. The primary mission focuses on dust sample return from Jupiter's Gossamer ring using aerogel-based collection technology. The secondary mission, assigned to a sub-probe, is dedicated to exploring Jupiter's Galilean moons. The probe executes multiple gravity-assist flybys of Europa and Callisto and performs close observations, enabling detailed investigation of their surface and near-space environments.

The remainder of this paper is organized as follows: Section 2 describes the scientific goals of the mission. Section 3 introduces the orbital theory used in the mission design, including Tisserand diagrams and the Multi-Gravity Assist and Deep Space Maneuver (MGA-DSM) model. Section 4 investigates the sample return mission design. The EVEEJEEME sequence is provided. Section 5 focuses on the trajectory design in the Jovian System. Section 6 draws the conclusion.

2. Scientific Goals

2.1 Sample Return Mission

Jupiter has a faint ring system comprising three main parts: the halo ring, the main ring, and the gossamer rings. The halo is the innermost part, extending about 123,000 km from Jupiter's center. It contains a higher concentration of fine dust and smaller particles. The main ring is the brightest and most well-defined component of its ring system. It primarily comprises micrometer-sized dust particles generated by micrometeoroid impacts on the small inner moons Adrastea and Metis. The outermost component is the Gossamer ring, extending from approximately 129,000 km to 250,000 km from Jupiter's center. It is formed from dust released by Amalthea and Thebe, as well as a result of micrometeoroid bombardment. Given that Jupiter's rings consist predominantly of fine dust, an aerogel-based sampling system—similar to that used in NASA's Stardust mission—is adopted for dust collection. The collected samples will be returned to Earth via a recovery capsule for detailed laboratory analysis^[4].

The target sampling region for this mission is Jupiter's Gossamer ring, located at an approximate radial distance of 240,000 km from the planet's center. Dust particles in this region typically range from 0.1 to 10 micrometers. They are influenced by perturbative forces, including Jupiter's strong magnetic and electric fields and solar radiation pressure. These forces result in complex, nonlinear dynamical behavior.

The sample return mission from the Gossamer ring will enable investigation into the surface weathering and dust ejection processes of Jupiter's moons. Moreover, the data will contribute to a deeper understanding of planetary ring formation and evolutionary mechanisms. Characterizing

the density and spatial distribution of micrometer-scale dust may provide valuable insights for designing protective dust shields and developing improved avoidance strategies for future spacecraft operating in dusty environments.

2.2 Exploration Targets within the Jovian System

According to the mission plan, following the completion of Jupiter exploration, the sample return module's engine will turn on at an appropriate window to enter a Jupiter-to-Earth return trajectory, delivering collected samples back to Earth. The sub-spacecraft will not return; instead, it will maneuver into an inner Jovian system orbit to continue extended scientific investigations of Jupiter and its four major Galilean moons.

This mission focuses on Callisto as the primary target for detailed observation, with planned gravity-assist flybys of Europa and Ganymede during the gravity-assist trajectory.

Callisto is of particular scientific interest due to its unique and complex characteristics. Its surface is believed to be covered by a thick ice shell, beneath which a subsurface saline ocean, potentially hundreds of kilometers deep, may exist, raising the possibility of habitable conditions. Despite its thin atmosphere, oxygen and ozone have been detected.

Callisto experiences a relatively lower radiation environment than Jupiter's inner moons, making it more suitable for long-term missions. Studying this environment deepens understanding of its physical and geophysical conditions and provides valuable insights into the hazards associated with deep-space missions. This information is crucial for evaluating the risks and developing protective strategies for future spacecraft and even human exploration into and beyond Jupiter.

3. Preliminary Dynamics Theory for Mission Design

3.1 Overview of Interplanetary Gravity Assist

When a spacecraft approaches a secondary celestial body in orbit around a primary gravitational source, the gravitational influence of the secondary body can alter its trajectory. In interplanetary missions, where the Sun serves as the primary body and planets as secondary bodies, this maneuver, known as a gravity assist or planetary flyby, modifies the spacecraft's heliocentric orbital energy.

The dynamics of gravity assist flight are governed by the relationship between the spacecraft's velocity relative to the assisting body and its velocity in the heliocentric inertial frame, as illustrated in Fig. 1. Upon entering the gravitational sphere of influence of the assisting body, the spacecraft possesses an incoming hyperbolic excess velocity $v_{\infty/in}$ relative to the body, and exits with an outgoing hyperbolic excess velocity $v_{\infty/out}$. The assisting body's heliocentric orbital velocity is denoted as v_{pl} . The spacecraft's entry and exit velocity vectors are separated by a deflection angle δ , which characterizes the turning effect induced by the flyby.

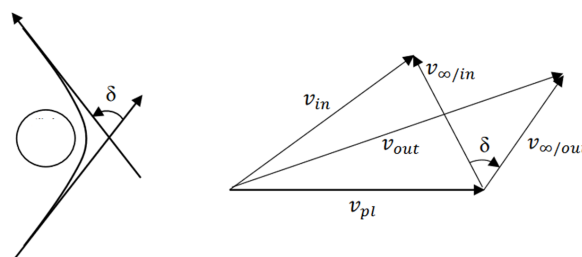


Figure 1. The Principle of Gravity Assist

Within the gravitational field of the assisting body, the spacecraft's asymptotic speed relative

to the body remains constant, i.e.,

$$\bar{\mathbf{o}} \mathbf{v}_{\infty/in} \bar{\mathbf{o}} = \bar{\mathbf{o}} \mathbf{v}_{\infty/out} \bar{\mathbf{o}} = \mathbf{v}_{\infty} \quad (1)$$

The direction of the velocity vector changes due to gravitational deflection. In the heliocentric inertial frame, the spacecraft's velocity before and after the flyby can be expressed as:

$$\mathbf{v}_{in} = \mathbf{v}_{pl} + \mathbf{v}_{\infty/in} \quad (2)$$

$$\mathbf{v}_{out} = \mathbf{v}_{pl} + \mathbf{v}_{\infty/out} \quad (3)$$

The resulting velocity increment $\Delta \mathbf{v}$ is defined as the difference between the post- and pre-encounter velocities in the heliocentric frame:

$$\Delta \mathbf{v} = \mathbf{v}_{out} - \mathbf{v}_{in} \quad (4)$$

The magnitude of this increment depends on the angle between $\mathbf{v}_{\infty/in}$ and $\mathbf{v}_{\infty/out}$, and is given by:

$$\bar{\mathbf{o}} \Delta \mathbf{v} \bar{\mathbf{o}} = 2v_{\infty} \sin(2\delta) \quad (5)$$

The deflection angle δ is a function of the planetary mass M_{pl} , the spacecraft's periapsis distance r_p , and the hyperbolic excess velocity v_{∞} , and is given by^[5]:

$$\delta = 2 \arcsin \left(\frac{1}{1 + \frac{r_p v_{\infty}^2}{\mu}} \right) \quad (6)$$

From the above, it is evident that the key to gravity assist lies in utilizing the gravitational field of the assisting body to change the spacecraft's velocity in the heliocentric frame. Through proper design of the flyby parameters, it is possible to achieve significant adjustments in the spacecraft's velocity, thereby meeting the requirements of interplanetary mission design.

3.2 Tisserand Graph to Select the Planetary Flyby Sequences

The Tisserand graph is a practical and visual method for designing multiple gravity assist (MGA) trajectories. The tool is rooted in an equation the French astronomer Tisserand formulated to predict how a body's orbital elements change following a planetary flyby. This relationship, now known as the Tisserand criterion, is expressed as:

$$T = \frac{a_p}{a} + 2 \sqrt{\frac{a}{a_p} (1 - e^2)} \cos i \approx \text{Constant} \quad (7)$$

The criterion establishes a near-conserved quantity linking the orbital parameters—semimajor axis, eccentricity, and inclination—of a spacecraft or small body to those of a flyby planet, characterized by semimajor axis.

In the case of coplanar orbits ($i=0$), this equation allows for a graphical representation of orbits on a periapsis–apoapsis (r_p – r_a) plot. Each point on the graph corresponds to a unique orbit. Additionally, the graph can include contours representing orbits with a constant hyperbolic excess velocity relative to a planet. These v_{∞} contours—plotted as colored curves for each planetary body—collectively form what is known as the Tisserand graph. This visualization aids trajectory planners in identifying feasible sequences of flybys based on dynamic constraints.

On a Tisserand graph, intersections between contours of equal hyperbolic excess velocity v_{∞} represent orbital configurations that enable a spacecraft to transfer between planets. Each flyby alters the spacecraft's trajectory by exploiting gravitational forces, yet conserves its v_{∞} magnitude relative to the flyby planet. Consequently, both planetary flybys and deep space maneuvers appear as transitions along the same v_{∞} contour.

For the Earth-to-Jupiter transfer orbit design, the candidate planetary bodies considered for

gravity assists include Venus (2), Earth (3), Mars (4), and Jupiter (5). The designated launch date is December 2029, with constraints on the hyperbolic excess velocity at departure and arrival: $3\text{km/s} < v_{\text{dep_inf}} < 6\text{km/s}$ at Earth departure, and $6\text{km/s} < v_{\text{arr_inf}} < 12\text{km/s}$ at Jupiter arrival. The corresponding Tisserand graph for Venus, Earth, Mars, and Jupiter is presented in Fig. 2.

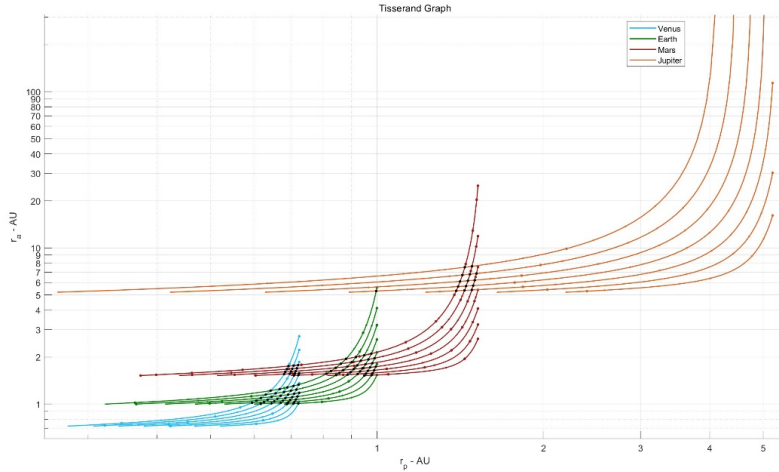


Figure 2. Tisserand graph

The graph identifies feasible planetary flyby sequences based on the connectivity of the Tisserand curves. The number of gravity assists is typically limited to one to three to avoid excessively long mission durations. Accordingly, three reasonable trajectory options derived from the graph include EMEJ, EVEEJ, and EMEEJ.

3.3 Interplanetary Transfers via Multiple Gravity Assists and Differential Evolution Optimization

In this paper, an interplanetary trajectory optimization framework is developed based on the combination of Multiple Gravity Assists (MGA) and Deep Space Maneuvers (DSM). The trajectory is modeled as a constrained nonlinear optimization problem to minimize the total velocity increment required throughout the mission. The decision vector is defined as:

$$\mathbf{y} = [t_0, v_\infty, \alpha, \delta, \{\eta_i\}, \{T_i\}, \{\gamma_i\}, \{R_i\}] \quad (8)$$

where t_0 denotes the launch epoch, v_∞ is the hyperbolic excess velocity magnitude at Earth departure, and α, δ represent the right ascension and declination of the departure asymptote, respectively. The set $\{\eta_i\}$ defines the time allocation ratio between the Kepler and Lambert phases for each transfer leg, $\{T_i\}$ are the time-of-flight durations between adjacent planetary encounters, $\{\gamma_i\}$ are the B-plane orientation angles for each gravity assist, and $\{R_i\}$ are the pericenter radius multipliers representing flyby altitudes.

The objective function incorporates both the total delta-v cost and soft constraint penalties:

$$J(\mathbf{y}) = \sum_{i=1}^N \Delta v_i + \rho_1 \cdot \max(0, -E) + \rho_2 \cdot \max(0, T_{\text{total}} - T_{\text{max}}) \quad (9)$$

where Δv_i is the maneuver cost of each segment, E is the orbital energy within the planetary sphere of influence (required to be negative for a valid gravity assist), and T_{total} is the cumulative flight duration, constrained by an upper bound T_{max} . The optimization problem is solved using the Differential Evolution algorithm, which is well-suited for non-convex global optimization in high-dimensional search spaces. This method outputs a feasible trajectory satisfying physical constraints, with applications to deep space missions requiring energy-efficient transfer designs.

4. Interplanetary Transfer and Sample Return Mission orbital design

Leveraging a trajectory optimized via a Multi-Gravity Assist and Deep Space Maneuver (MGA-DSM) model, the spacecraft follows an Earth–Venus–Earth–Earth–Jupiter (EVEEJ) path, enabling two crossings of the ring during the Jupiter flyby. This configuration provides favorable conditions for dust sampling. Upon completion of the sampling operation, the main probe separates and follows a Jupiter–Earth–Earth–Mars–Earth (JEEME) return trajectory. The collected samples will be returned to Earth for laboratory analysis, enabling studies on satellite-ring interactions, dust generation mechanisms, and the broader processes governing planetary ring formation and evolution, thereby contributing to foundational planetary science and supporting future deep space exploration.

4.1 Interplanetary Transfer Segment

Based on the preceding analysis, Considering the probe's launch date, the launch energy (C3), the arrival C3 at Jupiter, the magnitude of the deep-space maneuvers, and the overall mission duration, the EVEEJ (Earth–Venus–Earth–Earth–Jupiter) sequence is identified as the optimal trajectory.

The Earth departure window is scheduled for December 2029. To further optimize fuel efficiency, the mission design avoids applying a direct braking impulse upon arrival at Jupiter. Instead, a gravity assist maneuver at Jupiter is employed to adjust the spacecraft's trajectory in preparation for the subsequent capture phase. The detailed parameters of the corresponding transfer orbit are presented below.

Table 1. Summary of Interplanetary Pulse Orbit Parameters

	Date	Velocity Increment(km/s)	Flyby Radius(km)	v_{∞} (km/s)
Launch at Earth	2029/12/3	-	-	4.0152
DSM1	2030/1/9	0.0317	-	-
Flyby at Venus	2030/5/23	-	6678.3	7.2749
DSM2	2030/9/1	0.0323	-	-
Flyby at Earth 1	2031/4/9	-	9282.5	10.9438
DSM3	2031/11/29	0.0091	-	-
Flyby at Earth 2	2033/4/8	-	7854.6	10.9693
DSM4	2033/10/21	0.0115	-	-
Flyby at Jupiter	2035/9/26	-	240394.3	6.0806
Total Velocity Increment: 0.0845km/s				
Duration: 5.81 year				

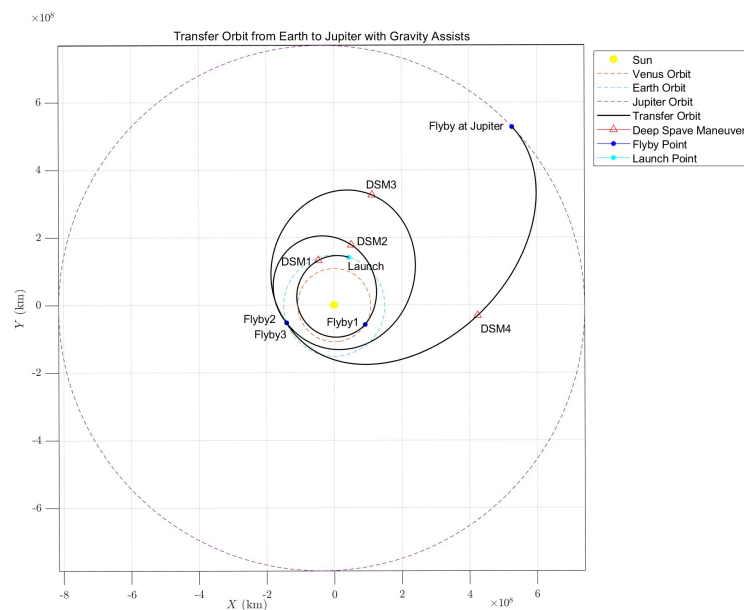


Figure 3. Transfer Orbit from Earth to Jupiter with Gravity Assists

4.2 Sample Collection Segment

The radial span of Jupiter's Gossamer ring extends from approximately 129,000 km to 250,000 km from Jupiter's center. According to the optimized MGA trajectory, the spacecraft's closest approach during the Jupiter flyby occurs at an altitude of 240,394 km, with a relative velocity v_{∞} of 6 km/s. This trajectory results in two crossings of the ring system—once during approach and once during departure—making it highly favorable for a ring sampling mission.

The sampling strategy is inspired by NASA's *Stardust* mission, which successfully collected dust from a comet's tail using aerogel at relative velocities exceeding 6 km/s. Given the comparable velocity in the proposed Jupiter flyby, it is technically feasible to adopt a similar method. Therefore, incorporating an aerogel-based sampling system as the spacecraft's primary payload for collecting material from Jupiter's rings is reasonable and supported by prior mission heritage.

4.3. Earth Return Segment

Upon completion of the sample-return mission, the sub-probe segment containing the return capsule will return to Earth at a suitable return window. The return trajectory is also designed using a multi-body gravitational assist strategy, enabling an efficient trajectory to minimize energy consumption. To ensure dynamical continuity, the return trajectory is initialized using the terminal position and velocity from the preceding Jupiter flyby, allowing a seamless transition in the mission profile.

The relevant orbital parameters are summarized in the following table and illustrated in the accompanying figure. The probe is expected to arrive back at Earth on September 27, 2044, marking the end of an 8.99-year mission.

Table 2. Optimal Trajectories of GA Sequences Returning to Earth

	Date	Velocity Increment(km/s)	Flyby Radius(km)	v_{∞} (km/s)
DSM1	2036/8/5	0.003	-	-
Flyby at Earth 1	2038/4/21	-	8422	9.8849

DSM2	2039/9/13	0.24	-	-
Flyby at Earth 2	2041/5/12	-	12588	7.6168
DSM3	2042/4/4	9.96e-4	-	-
Flyby at Mars	2043/6/10	-	16774	6.1522
DSM4	2044/2/13	0.0044	-	-
Arrival at Earth	2044/9/27	-	-	5.9829
Total Velocity Increment: 0.2483 km/s				
Re-entry Velocity: 12.589 km/s				
Duration: 8.99 year				

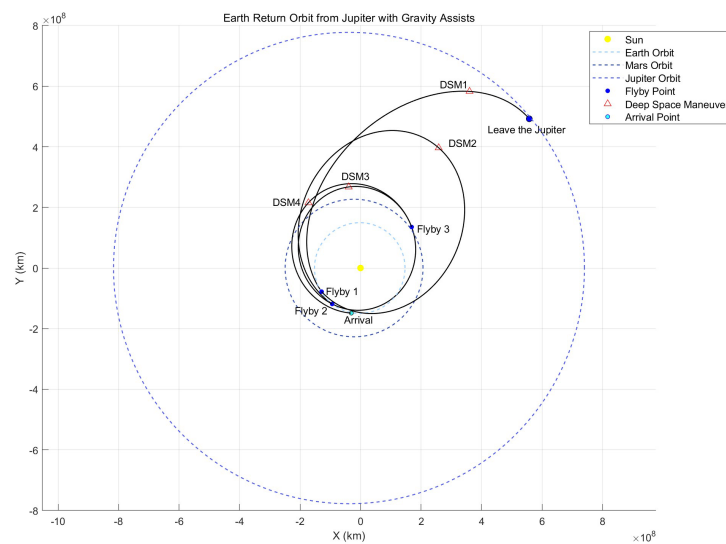


Figure 4. Transfer Orbit from Jupiter to Earth with Gravity Assists

Four deep-space maneuvers are executed along the return trajectory to correct accumulated deviations from the extended interplanetary cruise phase. These maneuvers refine the trajectory and ensure accurate targeting for Earth return. The spacecraft arrives at Earth with a v_{∞} of 5.982 km/s to enter the Earth's Sphere of Influence, and a re-entry speed of 12.589 km/s to enter the Earth's atmosphere.

The sample return mission requires a total velocity increment of 0.3329 km/s with a duration of 14.8 years. Fig.5 illustrates the complete trajectory of the sample return mission.

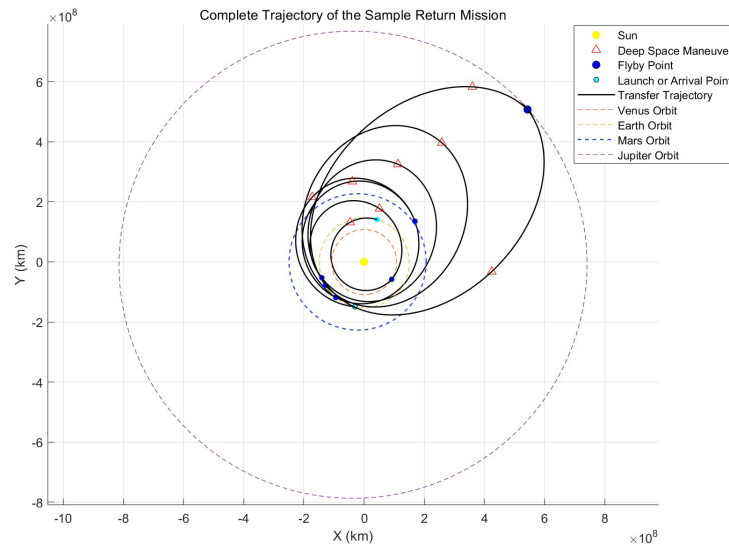


Figure 5. Complete Trajectory of the Sample Return Mission

5. Trajectory Design in the Jovian System

After separation from the main probe, the sub-probe initiates a series of maneuvers to enter the Jovian system, following an optimized trajectory that integrates planetary gravity assists, impulsive maneuver planning, and resonant orbit design. The mission architecture is divided into three key phases: Jupiter Capture, Jovian Gravity-Assisted Transfer, and Callisto Capture.

Throughout this sequence, Tisserand graph analysis, combined with the Multi-Gravity Assist and Deep Space Maneuver (MGA-DSM) optimization model, is employed to select and refine gravity-assist opportunities systematically. This approach ensures an efficient transfer from Earth to Callisto with minimized fuel consumption, achieving the required mission objectives while maintaining a minimal total ΔV within scientific and operational constraints.

5.1 Jupiter Capture Segment

Upon entry into the Jovian system, the Jupiter Capture orbit is designed using a high-fidelity orbital dynamics model that considers Jupiter's gravitational influence as the primary body and the perturbative effect of solar gravity. This model accurately represents the spacecraft's motion during the critical capture phase, especially given the complex gravitational environment near Jupiter^[6].

The dynamic equations for this segment, expressed in the Jupiter-centered rotating frame, are as follows:

$$\dot{\mathbf{r}} = \mathbf{v}$$

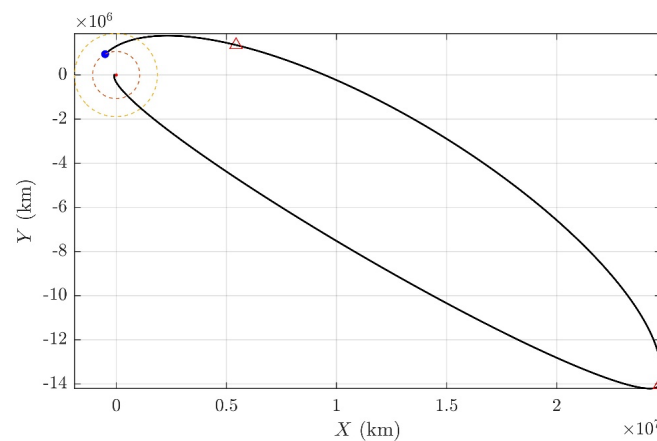
$$\dot{\mathbf{v}} = -\frac{\mu_{Jupiter}}{r^3} \mathbf{r} + \mu_{Sun} \left(\frac{\mathbf{r}_{spacecraft-Sun}}{r_{spacecraft-Sun}^3} - \frac{\mathbf{r}_{Jupiter-Sun}}{r_{Jupiter-Sun}^3} \right) \quad (10)$$

A Jupiter Orbit Insertion (JOI) impulse of 0.4186 km/s is applied at perijove to place the spacecraft into a highly elliptical orbit. Subsequently, a Perijove Raising maneuver (PJR) of 0.3793 km/s is executed at apojoive to increase the perijove altitude.

To further optimize conditions for upcoming gravity assists, a Deep Space Maneuver (DSM) of 0.0946 km/s is performed at approximately twice the orbital radius of Callisto. This segment requires a total ΔV of 0.8925 km/s and spans 339 days.

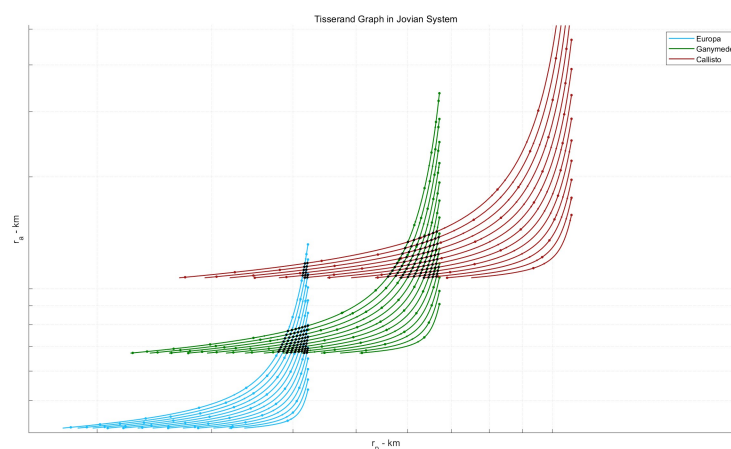
Table 3. Summary of Jupiter Capture Segment Impulse Maneuvers

Type of Pulse Maneuver	Delta v(km/s)
JOI	0.4186
PJR	0.3793
DSM	0.0946

**Figure 6.** Schematic of Jupiter's Capture Segment Orbit

5.2 The Jovian Gravity-Assisted Transfer Segment

The Tisserand diagram for the major moons of the Jovian system, as shown in Figure 7, is constructed to provide an initial assessment of feasible transfer opportunities. Based on the transfer parameters derived from these diagrams, the trajectory design is further refined using the Multi-Gravity Assist with Deep Space Maneuver (MGA-DSM) model under an actual ephemeris framework.

**Figure 7.** Tisserand Graph in Jovian System

The study uses the sequence of GGGGGEEEEEEGGGC to design a transfer orbit within the

Jovian system. This segment leverages multiple gravity assists from Ganymede to refine the spacecraft's orbit in two stages:

Stage 1 involves lowering the orbit via Europa and Ganymede flybys. The spacecraft utilizes a sequence of resonant encounters with Ganymede (9:1, 4:1, 3:1, and 2:1) to gradually reduce its orbital period from 316 days to 10 days, eventually enabling orbit intersection with Europa.

Stage 2 is the raising and pericenter adjustment of the orbit. A combination of resonant interactions with Europa (8:3, 7:3, 2:1, 5:3, 3:2) and further Ganymede encounters (3:4, 1:1) modifies the orbit to match the altitudes of Callisto orbits. This phase increases the orbital period from 5 to 11.5 days, aligning the spacecraft's trajectory with Callisto's orbital path.

Throughout this segment, orbital transitions are achieved with a total ΔV of 0.023 km/s over 348 days. The spacecraft arrives at Callisto with a relative hyperbolic velocity of 1.23 km/s.

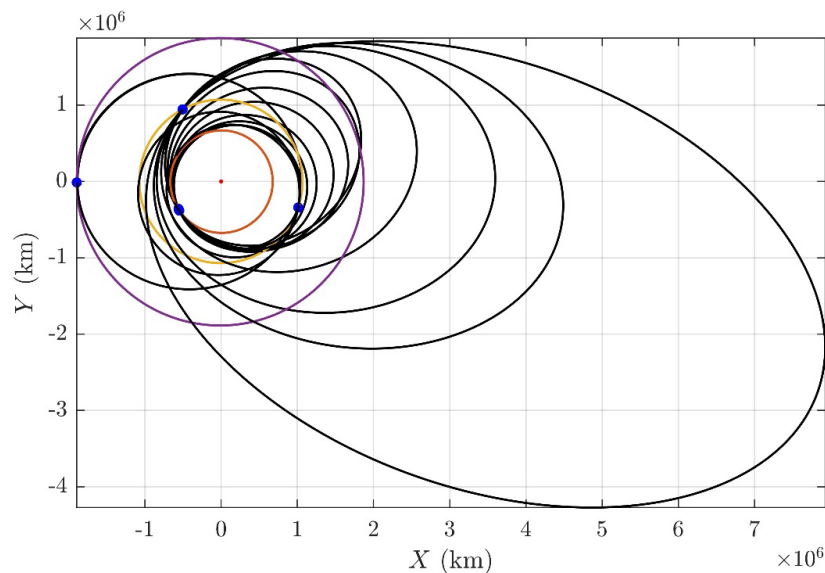


Figure 8. Schematic of Jovian Gravity-Assisted Transfer Orbit

5.3 Callisto Resonance Capture Segment

The Callisto Resonance Capture Segment (IRCS) reduces the relative hyperbolic velocity to Callisto through deep-space maneuvers and a final impulsive burn. Initial orbital tuning uses a 5:7 resonance ($\Delta V = 0.02$ km/s) followed by a 3:4 resonance ($\Delta V = 0.033$ km/s). Subsequently, a capture impulse of 0.8527 km/s is applied at perijove to transition the probe into a circular orbit at 300 km above Callisto's surface.

This segment lasts 131 days and requires a total ΔV of 0.9057 km/s, successfully achieving a stable orbit around Callisto.

5.4 Summary and Analysis

The entire transfer mission from Earth to the Jovian system spans 2940 days or 8.05 years (818 days within the Jovian System), with a total velocity increment of 1.9057 km/s (1.8212 km/s within the Jovian System). Fuel efficiency is achieved through each phase: the interplanetary transfer minimizes fuel use via planetary gravity assists; the Jupiter Capture segment applies a three-body dynamics model and leverages Ganymede's gravity for initial orbit insertion; the Jovian

gravity-assisted transfer adjusts the orbit using a sequence of resonances and multiple moons' flybys; and the Callisto capture phase combines resonant trajectories with targeted impulse maneuvers to ensure low-energy orbit insertion around Callisto.

Table 4. Summary of the Orbits within the Jovian System

Orbital Segment	Transfer Duration(day)	Delta v (km/s)
Jupiter Capture Segment	339	0.8925
Jovian Flyby Transfer Segment	348	0.0230
Callisto resonance capture segment	131	0.9057
Total	818	1.8212

This study presents a preliminary design and optimization of the Jupiter intragalactic probe orbit. Fig.9 illustrates the schematic of the entire mission.

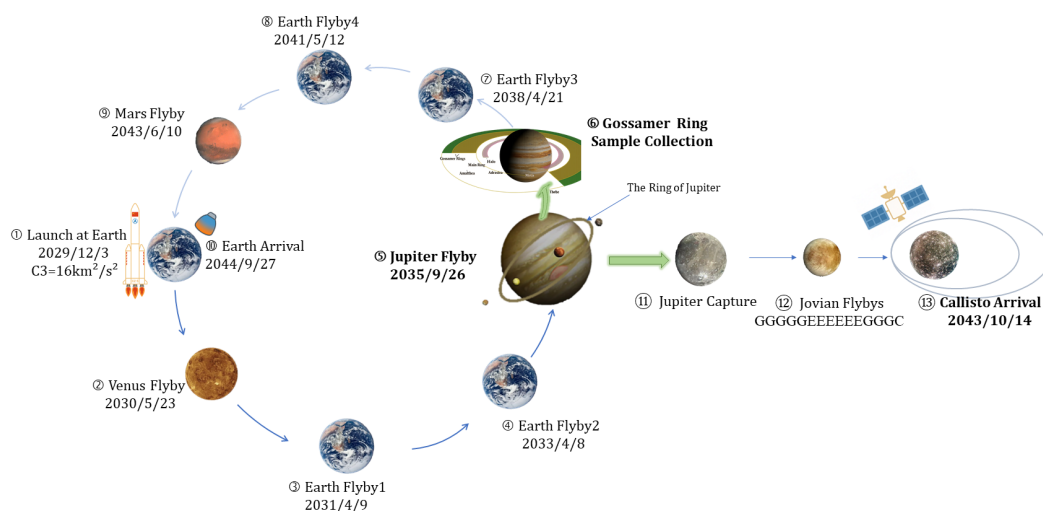


Figure 9. Schematic Diagram of the Mission Process

Significant potential remains for reducing the total fuel consumption through refined engineering approaches. In particular, during the Jupiter Orbit Insertion (JOI) phase, several strategies may be employed to reduce the required velocity increment:

(a) Electrodynamic Tether Capture (ETC): This approach exploits Jupiter's strong magnetic field. A conductive fluid tether can convert kinetic energy into electrical energy^[7], enabling passive, fuel-free braking and orbital capture.

(b) Solar Electric Propulsion (SEP): In the high solar radiation environment near Jupiter, SEP systems can provide sustained low-thrust maneuvering within the planet's strong magnetic field, thereby decreasing reliance on chemical propulsion^[8].

In addition to minimizing fuel use during capture, further reductions can be achieved during transfers within the Jovian system. The Laplace resonance involving Io, Europa, and Ganymede offers natural dynamical structures that can be harnessed for lower fuel consumption transfers. Moreover, transfer trajectories that require minimal propulsion can be identified by employing

dynamics techniques such as invariant manifold theory and low-energy transfer corridors. These methods allow the probe to navigate between multiple targets within the Jovian system with significantly reduced fuel consumption, enhancing the mission's scientific reach without exceeding launch mass constraints.

Several limitations remain. The influence of Jupiter's intense radiation environment on spacecraft systems and mission longevity must be more rigorously quantified. Ongoing work will investigate low-energy transfer options within the Jovian system to further reduce propellant consumption and enhance mission feasibility.

6. Conclusion

This study presents an innovative program that proposes an optimized orbital design for a multi-target exploration mission within the Jupiter system. It focuses on a Gossamer ring sample return objective, alongside flybys and orbital surveys of Jupiter and its major moons. Utilizing a multi-gravity assist (MGA) strategy in combination with a differential evolution optimization algorithm, the EVEEJ (Earth–Venus–Earth–Earth–Jupiter) sequence is identified as the most efficient interplanetary transfer trajectory. This path enables two crossings of Jupiter's Gossamer ring during the Jupiter flyby, creating a viable opportunity for dust sample collection using aerogel-based capture technology. The collected samples are returned through the JEEME (Jupiter–Earth–Earth–Mars–Earth) sequence, optimizing the mission's overall ΔV and timeline.

For the secondary target within the Jovian system, the mission design incorporates the Jupiter capture, followed by a series of gravity-assisted orbital transfers involving multiple flybys of Europa and Ganymede, culminating in orbital insertion around Callisto. This trajectory supports repeated flybys and orbital exploration of the Galilean moons, enabling broad scientific observations. The study may serve as a valuable reference for future Jupiter exploration mission design.

Acknowledgment

This paper is supported by the National Natural Science Foundation of China (No. 12150008, No. 62394353) and the Beijing Institute of Technology Research Fund Program for Young Scholars (Grant No. XSQD-202101012).

References

- [1] Zou X, Peng J and Miao Y M 2023 Research and prospect of scientific exploration for Jovian system *Chin. Space Sci. Technol.* **43**(6) 1–10
- [2] Gao B, Chen Z and Zhou W 2021 Progress and analysis of foreign Jovian exploration missions *Spacecraft Engineering*. **30**(5) 107–114 <https://doi.org/10.3969/j.issn.1673-8748.2021.05.016>
- [3] NASA 2025 *Juno Mission Overview* (Washington, DC: NASA) [online] Available at: <https://science.nasa.gov/mission/juno/> [Accessed 2 April 2025]
- [4] NASA 2025 *Stardust / Stardust NExT* (Washington, DC: NASA) [online] Available at: <https://science.nasa.gov/mission/stardust/> [Accessed 3 April 2025]
- [5] Ma W 2006 *Trajectory design and optimization for deep space exploration probe* MSc Thesis Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences, Beijing, China
- [6] Lynam A E, Kloster K W and Longuski J M 2011 Multiple-satellite-aided capture trajectories at Jupiter using the Laplace resonance *Celest. Mech. Dyn. Astron.* **109**(1) 59–84 <https://doi.org/10.1007/s10569-010-9307-1>
- [7] Schadeegg M M, Russell R P and Lantoine G 2015 Jovian orbit capture and eccentricity reduction using electrodynamic tether propulsion *J. Spacecr. Rocket.* **52**(2) 506–516 <https://doi.org/10.2514/1.A32962>
- [8] Strange N, Landau D, Hofer R, Snyder J, Randolph T, Campagnola S, Szabo J and Pote B 2012 Solar Electric Propulsion Gravity-Assist Tours for Jupiter Missions *AIAA/AAS Astrodynamics Specialist Conference* (Minneapolis, MN, August 13–16) <https://doi.org/10.2514/6.2012-4518>